Ananya Joshi

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Experience

Commodity and Ingredient Hedging August 2023 – Present
Commodity risk management technology serving 10,000+ agricultural and financial clients. *Chicago, IL*

Distributed Processing & Python Services | *Python, FastAPI, Redis/RQ, AWS, MySQL*

- Built distributed Python services processing 10K+ daily calls, using Redis/RQ to coordinate concurrent workers and MySQL to maintain durable workflow state across ingestion, file transfer, and archival stages
- Designed concurrency and fault-tolerance mechanisms using transactional row locking, idempotent processing, automatic retries, persisted failure states, and dedicated recovery workflows
- Developed FastAPI-based asynchronous job infrastructure to decouple API requests from background workloads while maintaining durable execution history, operational monitoring, and automated failure notifications
- Integrated AWS S3 and Bedrock into production data workflows, building AI-powered transcript processing and relational models that reduced manual client note-taking by 80%

Commodity Market Data & Analytics | *Python, SQL, AWS*

- Built an automated Python data ingestion pipeline for USDA WASDE reports, retrieving, versioning, and processing commodity supply-and-demand data from external sources
- Designed relational data models and idempotent processing workflows to track report releases, source-file versions, and ingestion state, enabling reliable processing of revised market data

Full-Stack Platform Modernization | *React, TypeScript, REST APIs*

- Led the architectural pitch and ongoing migration of internal commodity risk-management applications from legacy CFML to a decoupled React and REST API architecture
- Built a modular UI library and reusable React hooks with TanStack Query to standardize forms, server-state management, and API interactions, reducing frontend boilerplate by 30%
- Established strongly typed frontend models and standardized backend repository patterns to improve consistency across full-stack feature development

Internal Analytics & Business Intelligence | *SQL, Lucee, JavaScript*

- Owned backend development for an internal analytics platform, translating business requirements into data models and aggregation workflows spanning client interactions, trades, visits, tasks, and calendar activity
- Designed historical analytics models linking client activity and revenue, enabling business users to drill into client performance, analyze trends over time, and track changes across reporting periods
- Extended the analytics architecture from individual reporting to hierarchical team analytics, enabling leaders to aggregate employee metrics and evaluate performance across their organizations

Personal Projects

AI-Assisted Pixel Art Agent | *Python, TypeScript, Aseprite API, LLMs*

- Built an agent-driven workflow that programmatically interacts with Aseprite assets, enabling natural-language creation and modification of sprites, layers, and animations
- Developed an Aseprite scripting interface and tool-calling architecture to expose editor operations to an LLM while preserving structured project state

Education

University of Michigan Ann Arbor, MI
Bachelor of Engineering in Computer Science *Sept. 2019 – May 2023*

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C/C++
Backend & Data: FastAPI, Redis, RQ, SQLAlchemy, SQLModel, MySQL, REST APIs
Frontend: React, TanStack Query, HTML/CSS, Highcharts
Cloud & Infrastructure: AWS (S3, Bedrock), Docker, Git